E5. ESTRAZIONE DEL DNA DA TESSUTO DI MAMMIFERO

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BIBLIOGRAFIA

Current Protocols in Molecular Biology
(https://hoclai.wordpress.com/wp-content/uploads/2015/09/current-protocols-in-molecular-biology.pdf)

INTRODUZIONE

Tissue is rapidly frozen and crushed to produce readily digestible pieces. The processed tissue is placed in a solution of proteinase K and SDS and incubated until most of the cellular protein is degraded. The digest is deproteinized by successive phenol/chloroform/isoamyl alcohol extractions, recovered by ethanol precipitation, and dried and resuspended in buffer.

E5.1 REAGENTI

- **Tissues**

- **EDTA** (ethylenediamine tetraacetic acid) 0.5 M (pH 8.0)
  Dissolve 186.1 g $Na_2EDTA \cdot 2H_2O$ in 700 ml H_2O
  Adjust pH to 8.0 with **10 M NaOH** (~50 ml)
  Add H_2O to 1 liter

- **Tris·Cl** [tris(hydroxymethyl)aminomethane], 1 M
  Dissolve 121 g Tris base in 800 ml H_2O
  Adjust to desired pH with concentrated HCl
  Mix and add H_2O to 1 liter
  Approximately 70 ml of HCl is needed to achieve a pH 7.4 solution, and approximately 42 ml for a solution that is pH 8.0.
  IMPORTANT NOTE: The pH of Tris buffers changes significantly with temperature, decreasing approximately 0.028 pH units per 1°C. Tris-buffered solutions should be adjusted to the desired pH at the temperature at which they will be used. Because the pKa of Tris is 8.08, Tris should not be used as a buffer below pH ~7.2 or above pH ~9.0.

- **Digestion buffer**
  100 mM NaCl
  10 mM Tris·Cl, pH 8
  25 mM EDTA, pH 8
  0.5% Sodium Dodecyl Sulfate (SDS)
  0.1 mg/ml proteinase K
  Store at room temperature
  The proteinase K is labile and must be added fresh with each use.

NOTA: l'EDTA inibisce gli enzimi cellulari che potrebbero degradare il DNA, mentre il SDS è un detergente che, rimuovendo i lipidi dalle membrane cellulari, facilita la lisi delle cellule.

- **PBS, ice cold**
  PBS (Phosphate-Buffered Saline)
  10× stock solution, 1 liter:
  80 g NaCl
  2 g KCl
  11.5 g Na2HPO4·7H2O
  2 g KH2PO4
  Working solution, pH ~7.3:
  137 mM NaCl
  2.7 mM KCl
  4.3 mM Na2HPO4·7H2O
  1.4 mM KH2PO4

- **7.5 M ammonium acetate**

- **70% and 100% ethanol**

- **TE (Tris/EDTA) buffer**, pH 8
  10 mM Tris·Cl, pH 7.4, 7.5, or 8.0 (or other pH; see recipe)
  1 mM EDTA, pH 8.0

- **Incubator or water bath at 50°C, with shaker**

- Additional reagents and equipment for trypsinizing adherent cells (APPENDIX 3F)

- **Phenol/chloroform/isoamyl alcohol extraction**

E5.1.1 PREPARATION OF BUFFERED PHENOL AND PHENOL/CHLOROFORM/ISOAMYL ALCOHOL

For some purposes, fresh liquefied phenol (88% phenol) can be used without further purification. However, for purification of DNA prior to cloning and other sensitive applications, phenol must be redistilled before use, because oxidation products of phenol can damage and introduce breaks into nucleic acid chains. Redistilled phenol for use in nucleic acid purification is commercially available, but must be buffered before use. Appropriately buffered phenol is also commercially available, but is somewhat more expensive and should not be stored for long periods of time (e.g., > 6 months).
CAUTION: Phenol can cause severe burns to skin and damage clothing. Gloves, safety glasses, and a lab coat should be worn whenever working with phenol, and all manipulations should be carried out in a fume hood. A glass receptacle should be available exclusively for disposing of used phenol and chloroform.

- **8-hydroxyquinoline**

- **Liquefied phenol**
- **50 mM Tris base** (tris(hydroxymethyl)aminomethane, unadjusted pH ~10.5)

- **50 mM Tris·Cl** [tris(hydroxymethyl)aminomethane], pH 8.0, 1 M
  Dissolve 121 g Tris base in 800 ml H2O
  Adjust to desired pH with concentrated HCl
  Mix and add H2O to 1 liter
  Approximately 70 ml of HCl is needed to achieve a pH 7.4 solution, and approximately 42 ml for a solution that is pH 8.0.
  IMPORTANT NOTE: The pH of Tris buffers changes significantly with temperature, decreasing approximately 0.028 pH units per 1°C. Tris-buffered solutions should be adjusted to the desired pH at the temperature at which they will be used. Because the pKa of Tris is 8.08, Tris should not be used as a buffer below pH ~7.2 or above pH ~9.0.

- **Chloroform**

- **Isoamyl alcohol**

1) Add 0.5 g of 8-hydroxyquinoline to a 2-liter glass beaker containing a stir bar.

2) Gently pour in 500 ml liquefied phenol or melted crystals of redistilled phenol (melted in a water bath at 65°C). The phenol will turn yellow due to the 8-hydroxyquinoline, which is added as an antioxidant.

3) Add 500 ml of 50 mM Tris base.

4) Cover the beaker with aluminum foil. Stir 10 min at low speed with magnetic stirrer at room temperature.

5) Let phases separate at room temperature. Gently decant the top (aqueous) phase into a suitable waste receptacle. Remove what cannot be decanted with a 25 ml glass pipet and a suction bulb.

6) Add 500 ml of 50 mM Tris·Cl, pH 8.0. Repeat steps 4 to 6 so that two successive equilibrations with Tris·Cl are performed, ending with removal of the second Tris·Cl phase.
The pH of the phenol phase can be checked with indicator paper and should be 8.0. If it is not, the Tris·Cl equilibration should be repeated until this pH is obtained.

7) Add 250 ml of 50 mM Tris·Cl, pH 8.0, or TE buffer, pH 8.0, and store at 4°C in brown glass bottles or clear glass bottles wrapped in aluminum foil. Phenol prepared with 8-hydroxyquinoline as an antioxidant can be stored ≤ 2 months at 4°C.

8) For use in DNA purification procedure, mix 25 vol phenol (bottom yellow phase of stored solution) with 24 vol chloroform and 1 vol isoamyl alcohol. Store up to 2 months at 4°C wrapped in foil or in a dark glass bottle.

NOTA: il fenolo/cloroformio/alcool isoamilico provocano la precipitazione delle proteine lasciando gli acidi nucleici (DNA ed RNA) in soluzione acquosa.

Use deionized, distilled water in all recipes and protocol steps.

E5.2 PREPARAZIONE DELLE CELLULE

Beginning with whole tissue:

1a. As soon as possible after excision, quickly mince tissue and freeze in liquid nitrogen. If working with liver, remove the gallbladder, which contains high levels of degradative enzymes.

2a. Starting with between 200 mg and 1 g, grind tissue with a prechilled mortar and pestle, or crush with a hammer to a fine powder (keep the tissue fragments, if crushing is incomplete).

3a. Suspend the powdered tissue in 1.2 ml digestion buffer per 100 mg tissue. There should be no clumps.

Lyse and digest cells

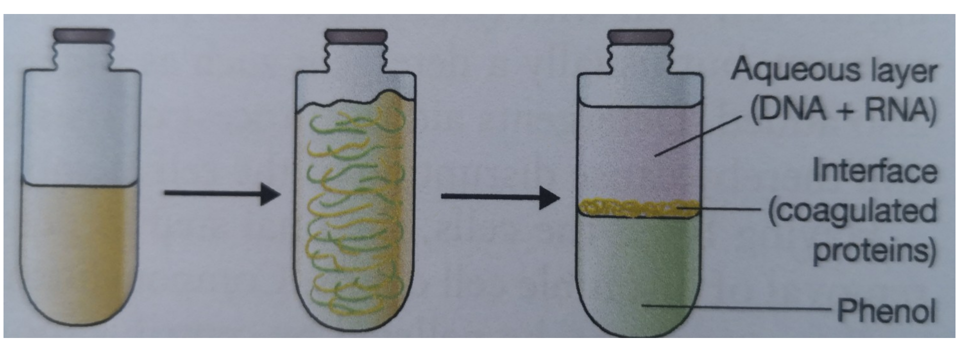
4. Incubate the samples with shaking at 50°C for 12 to 18 hr in tightly capped tubes. The samples will be viscous. After 12 hr incubation the tissue should be almost indiscernible, a sludge should be apparent from the organ samples.

E5.3 ESTRAZIONE DEGLI ACIDI NUCLEICI

5. Thoroughly extract the samples with an equal volume of phenol/chloroform/isoamyl alcohol.
CAUTION: Phenol is extremely caustic.

6. Centrifuge 10 min at 1700 × g in a swinging bucket rotor.

NOTA: Se l'estratto cellulare viene mescolato (piano) con il solvente, gli strati che si formano possono essere separati per **centrifugazione**; le molecole proteiche precipitate formeranno una massa bianca coagulata all'interfaccia tra la fase acquosa (contenente gli acidi nucleici in soluzione), in alto, ed il solvente, sotto (Figura E5.1). La soluzione acquosa degli acidi nucleici potrà, allora, essere rimossa con una pipetta.



Cell extract Mix with phenol Separate layers with centrifugation

Figura E5.1 Il fenolo provoca la precipitazione delle proteine.

If the phases do not resolve well, add another volume of digestion buffer, omitting proteinase K, and repeat the centrifugation. If there is a thick layer of white material at the interface between the phases, repeat the organic extraction.

E5.4 PURIFICAZIONE DEL DNA

7. Transfer the aqueous (top) layer to a new tube and add 1⁄2 vol of **7.5 M ammonium acetate** and 2 vol (of original amount of top layer) of 100% ethanol. The DNA should immediately form a stringy precipitate. Recover DNA by centrifugation at 1700 × g for 2 min. This brief precipitation in the presence of high salt reduces the amount of RNA in the DNA. For long-term storage it is convenient to leave the DNA in the presence of ethanol.

Alternatively, to prevent shearing of high-molecular-weight DNA, omit steps 7 to 9 and remove organic solvents and salt from the DNA by at least two dialysis steps against at least 100 vol TE buffer. Because of the high viscosity of the DNA, it is necessary to dialyze for a total of at least 24 hr.

8. Rinse (risciacquare) the pellet with 70% ethanol. Decant ethanol and air dry the pellet. It is important to rinse well to remove residual salt and phenol.

9. Resuspend DNA at ~1 mg/ml in TE buffer until dissolved. Shake gently at room temperature or at 65°C for several hours to facilitate solubilization. Store indefinitely at 4°C.
From 1 g mammalian cells, ~2 mg DNA can be expected.

If necessary, residual RNA can be removed at this step by adding 0.1% sodium dodecyl sulfate (SDS) and **1 μg/ml DNase-free RNase** and incubating 1 hr at 37°C, followed by organic extraction and ethanol precipitation, as above.
Ribonucleases (**RNases**) are used for a variety of analytical purposes, including hydrolyzing RNA that contaminates DNA preparations.

RIBONUCLEASE A

Ribonuclease A (RNase A) from bovine pancreas is an endoribonuclease that specifically hydrolyzes RNA after C and U residues. Cleavage occurs between the 3′-phosphate group of a pyrimidine ribonucleotide and the 5′-hydroxyl of the adjacent nucleotide. The reaction generates a 2′:3′ cyclic phosphate which then is hydrolyzed to the corresponding 3′-nucleoside phosphates.

Reaction Conditions

At NaCl concentrations of 0.3 M or above, RNase A becomes specific for cleavage of single-stranded RNA. Removal of RNase A from a reaction solution generally requires treatment with proteinase K followed by multiple phenol extractions and ethanol precipitation.

DNase-Free RNase A

To prepare RNase A free of DNase, dissolve RNase A in TE buffer at 1 mg/ml, and boil 10 to 30 min. Store aliquots at −20°C to prevent microbial growth.